Material Behavior Prediction

Portfolio Capstone | All 9 Milestones Complete

Executive Engineering Report & End-to-End Machine Learning Portfolio

Material Formulation (Composition) + Curing Condition (Time) → Machine Learning → Predicted Compressive Strength (MPa)

1. Executive Summary & Scientific Problem

In civil and materials engineering, concrete is the most consumed synthetic material on Earth (over 30 billion tons poured annually). Designing concrete formulations with target mechanical performance traditionally relies on trial-and-error laboratory batching: cylinders are poured and cured for **7, 28, or up to 365 days** before being destructively crushed in hydraulic frames.

Project Goal: Train, evaluate, interpret, and deploy machine learning models capable of estimating compressive strength directly from raw ingredient masses and curing age, dramatically accelerating the development of sustainable, low-carbon materials.

2. Dataset Provenance & Clean Architecture

We utilized the peer-reviewed benchmark dataset from **Prof. I-Cheng Yeh (1998)** published in *Cement and Concrete Research*:

- **Row Representation:** One physical test cylinder mixed and crushed in a certified materials laboratory.
- **Total Samples:** 1,030 experimental formulations (cleaned to 1,005 after removing exact replicates).
- **Input Features (8):** `cement, slag, ash, water, superplastic, coarseagg, fineagg, age` .
- **Target Output:** `strength` (Compressive Strength in Megapascals, continuous regression).
- **Hygiene:** Zero missing values across all columns; all values verified within physical boundaries (≥ 0).

3. The 4-Algorithm Benchmark Comparison (Milestone 5)

Evaluated on the strictly held-out **201 unseen test samples** (20% partition, `random_state=42`):

Model Architecture	Train R^2	Test R^2	Test MAE	Test RMSE	Generalization Gap	Engineering Assessment
Linear Regression (OLS)	0.6099	0.5802	8.90 MPa	11.19 MPa	0.0297	Underfits non-linear physics (Abrams' Law). Baseline benchmark.
Decision Tree Regressor	0.8641	0.7885	5.95 MPa	7.94 MPa	0.0755	Partitions physical regimes, but step boundaries lack smoothness.
Random Forest (100 trees)	0.9824	0.9069	3.55 MPa	5.27 MPa	0.0755	Averages bagging trees; cuts error by 60% compared to baseline.
Gradient Boosting ★	**0.9699**	**0.9198**	**3.45 MPa**	**4.89 MPa**	**0.0501**	**CHAMPION MODEL: Explains 92.0% variance with lowest error.**

Core Performance Achievement

Moving from the linear baseline to Gradient Boosting slashed prediction error from **8.90 MPa to 3.45 MPa** (a **61.2% error reduction**) and elevated explained variance to **92.0%** with minimal overfitting (gap = 0.05).

4. Model Interpretation: Combining ML with Materials Science (Milestone 6)

We probed the champion model using both **Tree MDI Importance** and **Test Set Permutation Importance**:

Feature	MDI Importance	Test Permutation Drop (Δ R^2)	Scientific & Chemical Mechanism
1. age	**37.3%**	**-0.7147**	Hydration kinetics: time required for crystalline C-S-H growth.
2. cement	**28.9%**	**-0.5357**	Primary hydraulic binder: raw chemical reactant forming solid matrix.
3. water	**9.6%**	**-0.1800**	Abrams' Law: governs capillary porosity; excess water creates weak voids.
4. superplastic	8.9%	-0.0722	Admixture: reduces water demand while maintaining fluidity.
5. slag	7.9%	-0.1436	Supplementary binder: secondary latent hydraulic reaction.
6. fineagg	4.6%	-0.0441	Microscopic skeleton: sand filling gaps between crushed stones.
7. coarseagg	1.6%	-0.0128	Macroscopic skeleton: bulk stone resisting mechanical deformation.
8. ash	1.2%	-0.0028	Pozzolanic binder: contributes to long-term strength past 56 days.

The Discovery of "The Big Three"

Together, `age` **(37.3%) +** `cement` **(28.9%) +** `water` **(9.6%) govern 75.8% of compressive strength**.
The machine learning model independently verified the two foundational pillars of concrete engineering:

1. **Abrams' Law (1918):** Strength decreases non-linearly with increasing water-to-cement ratio (w/c).
2. **Logarithmic Hydration Kinetics:** Strength surges between 1 and 28 days and asymptotically levels off by 90–365 days.

5. Physical Tolerance Benchmarks & Error Post-Mortem (Milestone 7)

- **Within ±2.5 MPa: 49.8%** of test formulations (tight laboratory gauge precision).
- **Within ±5.0 MPa: 75.6%** of test formulations (ASTM C39 standard batch-to-batch field acceptance).
- **Within ±10.0 MPa: 95.5%** of test formulations (broad structural safety envelope).

Transparent Limitation Identified: The largest errors occur exclusively in the ultra-high strength regime (>70 MPa), where the model underpredicts because high-strength batches represent only ~5% of training samples. Terminal tree leaf averaging prevents extrapolation into extreme upper tails.

6. New Material Inference & Deployment (Milestone 8)

The deployed `predict_material(...)` pipeline in `src/predict_new_material.py` validates physical bounds, formats feature schema, loads `models/final_material_model.joblib`, and outputs predicted strength alongside civil engineering application grades:

- **Standard Residential Mix (28 days, $w/b = 0.60$):** Predicted **31.31 MPa** (Standard Structural Grade).
- **Green High-Performance Mix (56 days, $w/b = 0.38$, slag + ash):** Predicted **57.08 MPa** (High-Performance Grade).
- **Rapid Early-Strength Repair (3 days, $w/b = 0.28$):** Predicted **38.39 MPa** (Heavy Commercial Grade).

7. Future Roadmap & Scientific Disclaimers

- **Next Steps:** Streamlit web interface with interactive sliders, SQL database for batch tickets, and Power BI / Tableau dashboard integration.
- **Scientific Integrity:** Model predictions are *statistical estimates* derived from empirical data and must be confirmed by standard destructive compression testing (ASTM C39) before life-critical deployment.